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ORIGINAL ARTICLE

Knowledge transfer into sport practice: An empirical user analysis of a sport science website

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Abstract
Transferring scientific knowledge to applied practitioners presents a challenge in sport science. Knowledge transfer, conceptualized as knowledge reuse and knowledge contribution, has mainly been studied in the context of specific topics, projects, or organizations. The purposes of this study were to investigate the determinants of knowledge transfer from a publicly available website to its users and to cluster users into different groups based on their level of knowledge transfer. The website www.sportsandscience.de is dedicated to translating evidence-based knowledge that is regularly generated in scientific studies into short and digestible information through e.g. videos and blogs, and provides opportunities for knowledge reuse and contribution. A theoretical model of knowledge transfer in organizations which was adapted to this context guided the study. Data from an online survey of registered website users were employed for the empirical analysis. The regression analyses showed that knowledge reuse is positively affected by ease of knowledge access and negatively by age. Knowledge reuse, job experience, and male gender have a significant positive effect on knowledge contribution. The findings suggest that the theoretical model is generally applicable to this context, but also highlight differences to the original organizational setting. Cluster analysis yielded three clusters: Frequent users, occasional users, and frequent contributors. Analyses of variances revealed significant differences among clusters regarding computer self-efficacy, job experience, gender, age, employment, interested sports, and conditional skills, but not concerning education and formal qualifications. Implications for improving knowledge transfer of such a website are provided.

Keywords: Education, management, technology

Highlights
- Study examines knowledge transfer from a sport science website based on data from a user survey.
- A theoretical model of knowledge transfer in organizations is applied to this context.
- Determinants of knowledge reuse and knowledge contribution are identified.
- Cluster analysis identified 3 clusters: Frequent users, occasional users, and frequent contributors.

Introduction

Knowledge transfer is defined as the process through which individuals or groups are influenced by the experiences of other individuals or groups (Argote & Ingram, 2000). It consists of two components (Watson & Hewett, 2006): *Knowledge reuse*, referring to the application of existing knowledge, and *knowledge contribution*, which implies that users also contribute their own knowledge to the existing body of knowledge, suggesting that knowledge transfer is a two-directional process.

Previous sport science research has examined knowledge transfer, though mainly knowledge reuse, within specific projects or topics such as health (Rütten & Gelius, 2012), elite sport (Rütten, 2007), gender (Hartmann-Tews & Dahmen, 2007), or sport medicine (Provvidenza & Johnston, 2009); within organizations such as sport governing bodies (O'Reilly & Knight, 2007) or professional team sport providers (Erhardt, Martin-Rios, & Harkins, 2014); and at sport events (Parent, MacDonald, & Goulet, 2014). The reuse of scientific knowledge by

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various sport practitioners such as policymakers (Rütten & Gelius, 2012), employees of sport governing bodies (O'Reilly & Knight, 2007), athletes (Gerbing & Thiel, 2016), and coaches (Reade, Rodgers, & Spriggs, 2008b) represents a challenge already.

For example, research on coaches suggests that knowledge reuse is hampered by the accessibility and comprehensibility of scientific knowledge: Coaches reported language and access barriers (Martindale & Nash, 2013) and lack of time to search for and digest knowledge and to physically attend coaching conferences or meet other coaches (Reade et al., 2008b). Overall, they regarded scientific knowledge as too abstract, because they prefer receiving specific leads and strategies (Reade, Rodgers, & Hall, 2008a). Coaches indicated that they would use scientific knowledge more frequently if provided in a more comprehensible and digestible manner (Pope et al., 2015); otherwise they consider it irrelevant for their work (Martindale & Nash, 2013). However, bridging the gap between scientists and practitioners by effectively transferring knowledge generated in scientific studies to applied practice and encouraging knowledge contribution is necessary to facilitate positive changes within all areas of sport.

Typically, scientific publications and conference presentations are mainly consumed by scientists, ensuring knowledge transfer only within the academic community. A more practical and applied way of providing scientific knowledge is web-based. One such example is the website www.sportsandscience.de which is dedicated to providing and translating evidence-based knowledge generated in scientific research into language understood by sport practitioners. The website has an editorial board of experts in different fields who refine new knowledge and provide meaningful information to applied practitioners. The editor scans social media contents on a daily basis and actively asks applied practitioners (e.g. coaches in different sports, governing bodies, physicians, etc.) for relevant topics where evidence-based knowledge is needed. Subject areas on the website include training and recreation, physical activity and health, diagnostics and monitoring, technology, environment, and nutrition.[1] The website represents a platform for knowledge reuse and contribution. Means for knowledge reuse include blog-style short articles, comments on scientific publications, blogs, podcasts, videos, webinars, online conferences, and presentation slides for teaching purposes. Knowledge contribution is possible by commenting on existing content.

Providing scientific knowledge through such a website may represent a mean to make academic knowledge accessible for sport practitioners and encourage knowledge contribution. However, the sport science community does not know enough about knowledge transfer from such a platform given that knowledge transfer, and predominantly knowledge reuse, have mainly been studied within the boundaries of specific projects, topics, or institutions. Therefore, the purpose of this study is to enhance our understanding of knowledge transfer based on the website www.sportsandscience.de. Specifically, this study advances two main research questions: (i) which factors affect the transfer of knowledge provided on a sport science website to the users of this website? And (ii) in what groups can website users be categorized and how do these groups differ from each other?

Theoretical model of knowledge transfer

This study applies the multi-theoretical model of knowledge transfer in organizations (Watson & Hewett, 2006), which is based on expectancy theory (Vroom, 1964) and social exchange theory (Blau, 1964). It also draws on the technology acceptance model (Davis, Bagozzi, & Warshaw, 1989; King & He, 2006) given its focus on knowledge transfer employing technology. Watson and Hewett (2006) propose several factors which positively affect the frequency of knowledge reuse and knowledge contribution. Figure 1 displays their original model and visualizes how it was adapted to fit the present research context where people work for different organizations or are self-employed.

Starting with knowledge reuse, expectancy theory assumes that individuals put forth effort to access knowledge when they believe it assists them in achieving their desired outcome and when they consider this outcome valuable (Vroom, 1964; Watson & Hewett, 2006). Ease of knowledge access is associated with higher expectations of obtaining relevant knowledge (Vroom, 1964) and thus affects people's intentions to access a knowledge system (Davis et al., 1989). Originally, training in knowledge use refers to informing employees about the sources of knowledge available to them and how they could be accessed (Watson & Hewett, 2006). Since the source of knowledge is fixed in this study (i.e. the website), this factor was adjusted to the training needed to understand and use the provided knowledge, also to be more distinct from ease of knowledge access. Computer self-efficacy reflects individuals' confidence and competency of using computers, and increases the likelihood that people perceive benefits of accessing knowledge and reusing it. Likewise, trust in the knowledge source in the sense that the knowledge provided is timely and accurate as

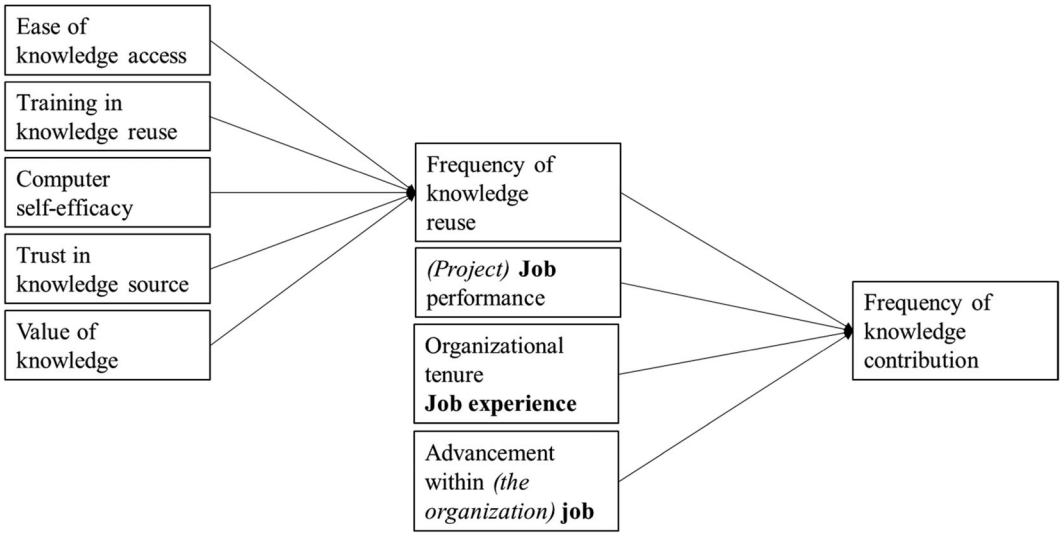

Figure 1. Theoretical model of knowledge transfer (slightly adapted from Watson & Hewett, 2006, p. 148). Note: Italicized words in parentheses are from the original version and were replaced by the bold words.

well as the value of the provided knowledge to individuals will enhance people's motivation to engage in knowledge reuse (Watson & Hewett, 2006).

Knowledge contribution is theoretically grounded in social exchange theory, stating that individuals who perceive to benefit from the provided knowledge will also contribute their valuable knowledge to bring this relationship into balance (Blau, 1964; Watson & Hewett, 2006). Individuals receiving a favour from another party in the form of reusing knowledge are likely to provide a service in return by making their own contribution to knowledge (Blau, 1964). In the absence of organizational boundaries, not all users of this study work on specific projects; hence project performance was changed to job performance which is relevant across employment forms. Organizational tenure affects the commitment to give something back which increases with an increasing number of years in an organization (Watson & Hewett, 2006). Job experience was added to the model to better capture job-specific human capital beyond organizational boundaries (Soebbing, Wicker, & Watanabe, 2016; Wicker, Orlowski, & Breuer, 2016). Advancement within the organization was changed to advancement within the job to cover all possible forms of employment of website users.

Method

Data collection

Users of the website www.sportsandscience.de were surveyed using an online survey. The investigation was performed in accordance with the Declaration of Helsinki and the ethical committee of the lead author's university. At the time of data acquisition, the website had approximately 2400 registered users who were invited via e-mail to participate in the survey on 21 March 2018 (150 e-mails bounced back). The e-mail informed users about the purpose of the study, that completion of the survey would take approximately 10 min, that the survey would be anonymous, and that the data would only be used for scientific purposes. On average, respondents needed 8 ± 2 min to fill in the survey. Users were also informed about the possibility to participate in the raffle of a Polar watch V800 after survey completion. This incentive was provided in an effort to increase the response rate. Three reminders were sent on 31 March 27 April, and 13 May 2018. The survey period ended on 16 May 2018.

Altogether, 443 users clicked on the survey link, 185 initiated, and 139 completed the survey, with the latter representing the final sample size for the statistical analysis. This sample size is smaller compared to previous studies on knowledge transfer (e.g. Reade et al., 2008a: $n = 205$; Watson & Hewett, 2006; $n = 430$), but larger than others (e.g. Gerbing & Thiel, 2016: $n = 110$; Martindale & Nash, 2013: $n = 58$; O'Reilly & Knight, 2007: $n = 25$; Reade et al., 2008b: $n = 20$). With about 6.2%, the completed response rate is lower than in other online surveys, typically yielding response rates of up to 30% (e.g. Van Deursen, van Dijk, & ten Klooster, 2015; Wicker et al., 2016). However, many online surveys were based on convenience and snowball sampling (e.g. Pope et al., 2015), implying that response rates could not be calculated because the total number of survey recipients was unknown.

Table I. Overview of variables and descriptive statistics

Variable	Description	Share of respondents	Mean	SD	Min	Max
Frequency knowledge reuse	Access of knowledge from sportsandscience.de at least once a month (1 = yes; 0 = no)	66.9%	–	–	0	1
Frequency knowledge contribution	Comment on articles/videos in a blog/forum on sportsandscience.de at least seldom (1 = yes; 0 = no)	18.7%	–	–	0	1
Ease of knowledge access	Difficulty of knowledge access from sportsandscience.de (1 = very difficult; 5 = very easy)	–	3.52	0.77	2	5
Training in knowledge reuse	Number of website articles for which specific training is needed to use the knowledge (1 = for every article; 5 = for no article)	–	3.55	0.78	2	5
Computer self-efficacy	Additive index based on 5 items (1 = not at all confident; 5 = very confident)	–	3.78	0.62	1.60	5.00
Trust in knowledge source	Level of trust in this website (1 = do not trust at all; 5 = trust highly)	–	4.44	0.67	2	5
Value of knowledge	Value of knowledge from sportsandscience.de to work (1 = not valuable; 5 = extremely valuable)	–	4.06	0.74	2	5
Performance	Contribution of knowledge provided on sportsandscience.de to job performance (1 = did not contribute at all; 5 = contributed greatly)	–	3.04	0.73	1	5
Organizational tenure	Number of years at current employer or self-employed	–	5.94	7.25	0	35
Job experience	Number of years in current job	–	8.42	8.37	0	35
Advancement within job	Career progression compared to peers/colleagues in this job (1 = much slower; 5 = much faster)	–	3.41	0.88	1	5
Knowledge level	Current level of knowledge in sport science (1 = very poor; 5 = very good)	–	3.83	0.89	1	5
Sport hours	Number of hours of sport per week	–	7.51	3.99	0	25
Male	Gender (1 = male; 0 = female)	77.7%	–	–	0	1
Age	Age (in years)		34.74	9.67	20	60

Questionnaire and variables

The survey included 36 questions and was equivalent to six pages in written form (12pt font). It was pre-tested among a handful of sport scientists and practitioners to ensure that the questions were comprehensible. The questionnaire consisted of three topics – usage habits on the website, knowledge transfer, and respondents' socio-demographics. Table I summarizes the resulting variables. Starting with usage habits, respondents were asked to provide information about the topics and sports they are interested in when they access knowledge on the website, the purposes they use the accessed knowledge for, and their current level of knowledge in sport science (*Knowledge level*).

The heart of the survey encompassed questions assessing the factors of the theoretical model of knowledge transfer in line with Watson and Hewett (2006). The wording of some questions had to be adapted to fit the present research context. Concerning knowledge transfer, respondents were asked to report how frequently they accessed knowledge from sportsandscience.de (*Frequency knowledge reuse;* several times a week: 2.2%; once a week: 14.4%; once a month: 50.4%; once in 6 months: 24.5%; less frequently: 8.6%) and how often they commented on articles/videos in a blog/forum on this website (*Frequency knowledge contribution;* almost always: 0%, frequently: 0%, sometimes: 4.3%; seldom: 14.4%; never: 81.3%). Since only few respondents were in the more frequent categories, the original five-point scale variables were recoded into dummy variables, with *1* reflecting at least once a month (knowledge reuse) and at least seldom (knowledge contribution), respectively.

Respondents were asked about the determinants of knowledge reuse, including the difficulty of accessing knowledge from the website (*Ease of knowledge access*), the extent to which they would need training to use the knowledge provided in articles (*Training in knowledge reuse*), how much they trust the website (*Trust in knowledge source*), and the value of the provided knowledge to their work (*Value of knowledge*). Concerning determinants of knowledge contribution, the contribution of the provided knowledge to their job performance (*Performance*), the number of years at their current employer or of being self-employed (*Organizational tenure*), the number of years in their current job (*Job experience*), and their career progression compared to peers/colleagues in this job (*Advancement within job*) were assessed.

Computer self-efficacy was measured using a scale in line with previous research (Compeau & Higgins, 1995; Watson & Hewett, 2006). Respondents were

asked to imagine how confident they would be to work with an unfamiliar software package and to indicate their degree of confidence on a five-point scale (1 = *not at all confident*; 5 = *very confident*). Since deploying new software is not only relevant to working with computers, but also other technological devices (e.g. smartphones, tablets), this scale should adequately capture the technical skills of users. The scale included the following five items: Someone gives me step by step instructions; No one is around to tell me what to do; I have never used a package like it before; I only have software manuals for reference; and someone else helps me get started. Cronbach's alpha of the scale is 0.692, indicating moderate reliability (Hinton, McMurray, & Brownlow, 2004). For unidimensional scales with less than 10 items like the present one, alphas from 0.6 onwards were regarded as acceptable (Hair, Black, Babin, & Anderson, 2010). The computer self-efficacy index was obtained by adding up the 5 items and dividing the sum by 5. The survey finished with a set of questions assessing respondents' socio-demographic characteristics, including gender (*Male*), age (*Age*), the number of weekly hours they practice sport (*Sport hours*), current job, educational level, formal sport- and health-related qualifications, and income.

Empirical analysis

The empirical analysis consisted of two parts to answer the two research questions. First, regression analyses were conducted to identify the determinants of knowledge transfer (Watson & Hewett, 2006). Two logistic regression models were estimated given the binary nature of the two dependent variables, *Frequency knowledge reuse* (Model 1) and *Frequency knowledge contribution* (Model 2). All variables from the theoretical model and selected socio-demographics (*Knowledge level, Sport hours, Male, Age*) were included as independent variables. Given the small sample size, the ratio between the number of respondents and the number of independent variables must be considered in regression analysis: Rules of thumb suggest ratios of 5:1 (Hair et al., 2010) or 2:1 (Backhaus, Erichson, Plinke, & Weiber, 2010); or a number of respondents = 50 + (8 + v), where v represents the number of independent variables (Tabachnick & Fidell, 2007). The present study meets each criterion. The independent variables were checked for multicollinearity using correlations analyses. Organizational tenure, job experience, and age were highly correlated, ultimately resulting in the former two variables being removed from Model 2.

Second, a two-step cluster analysis identified different user groups. The two knowledge transfer variables served as cluster variables in their ordinal scale. Analyses of variances (ANOVAs) were run to detect significant differences among clusters and, in the case of significant differences, a *post-hoc* test identified between which clusters the differences were statistically significant. ANOVAs were run for all variables of the two regression models and the variables capturing the topics or sports respondents were interested in, educational level, current job, formal sport- and health-related qualifications, and income. Given the large number of variables, only those variables will be reported where the ANOVA revealed significant differences among clusters.

Results

The summary statistics (Table I) show that respondents were mostly male (77.7%) and on average 34.7 years old. On average, they practiced sport for 7.5 h per week and considered their current level of knowledge in sport science as almost good (M = 3.83). Respondents have worked 8.4 years in their current job and 5.9 years with their current employer. Regarding knowledge reuse, 66.9% of respondents accessed knowledge from sportsandscience.de at least once a month, while 18.7% also contributed knowledge by commenting on articles/videos in a blog/forum.

Table II displays the results of the two regression analyses testing the theoretical model of knowledge transfer. The two models explained 16.4% (Model 1) and 20.4% (Model 2) of the variation in the dependent variable, respectively. In Model 1, ease of knowledge access had a significant positive effect on the frequency of knowledge reuse, while the effect of age was negative. Hence, older people were less likely to reuse knowledge. In Model 2, frequency of knowledge reuse, job experience, and male gender had a significant positive effect on the frequency of knowledge contribution, meaning that more frequent knowledge reusers, people with more job experience, and males were more likely to contribute knowledge.

Table III summarizes the results of the cluster analysis and the ANOVAs. The clustering procedure resulted in three clusters which were labelled frequent users (Cluster 1; 51.1%), occasional users (Cluster 2; 31.7%), and frequent contributors (Cluster 3; 17.3%). As indicated by the clusters' names, respondents in Cluster 2 used knowledge less frequently than those in Clusters 1 and 3, while those in Cluster 3 contributed knowledge more frequently than those in the first two clusters. Frequent

Table II. Results of logistic regression analyses

	Model 1: Frequency knowledge reuse	Model 2: Frequency knowledge contribution
Frequency knowledge reuse	–	1.975**
Ease of knowledge access	1.860**	–
Training in knowledge reuse	1.233	–
Computer self-efficacy	1.395	–
Trust in knowledge source	1.087	–
Value of knowledge	1.071	–
Performance	–	1.344
Job experience	–	1.048*
Advancement within the job	–	0.925
Knowledge level	1.105	1.375
Sport hours	0.995	1.025
Male	1.885	7.465*
Age	0.960*	–
Constant	0.031	0.000***
χ^2	17.396	18.757***
−2LL	159.090	115.216
Nagelkerke's R^2	0.164	0.204

Note: Displayed are the odds ratios; $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

users and contributors perceived the knowledge to be more easily accessible than occasional users. Computer self-efficacy was significantly higher among frequent users than in the other two clusters, while people in this cluster had significantly less experience in their job. The share of males was significantly higher among frequent contributors than among frequent and occasional users, while frequent users were significantly younger than people in the other two clusters.

Concerning the type of sports or conditional skills, frequent contributors were particularly interested in endurance, frequent users in strength, and occasional users in fitness and health. Frequent users were more interested in track and field, while occasional users were more interested in football. The analysis also revealed significant differences between clusters in terms of the respondents' job situation: People who were employed as academic coordinators, were self-employed, or employed in corporate health management were more likely to belong to the cluster of frequent contributors, while the share of students was highest among frequent users.

Table III. Results of the cluster analysis, ANOVA, and *post-hoc* test.

Variable	Cluster 1 (51.1%): Frequent users	Cluster 2 (31.7%): Occasional users	Cluster 3 (17.3%): Frequent contributors	*F*
Frequency knowledge reuse	3.24^{b}	1.73^{ac}	3.29^{b}	130.03***
Frequency knowledge contribution	1.00^{c}	1.05^{c}	2.25^{ab}	320.68***
Ease of knowledge access	3.65^{b}	3.25^{ac}	3.63^{b}	4.035**
Computer self-efficacy	3.91^{cb}	3.69^{a}	3.58^{a}	3.472**
Job experience	6.41^{bc}	10.41^{a}	10.71^{a}	4.397**
Male	0.761^{c}	0.705^{c}	0.958^{ab}	3.067**
Age	32.30^{bc}	37.36^{a}	37.17^{a}	4.912***
Sport: Endurance	0.085^{c}	0.046^{c}	0.208^{ab}	2.537*
Sport: Strength	0.310^{c}	0.182	0.042^{a}	4.195**
Sport: Fitness and health	0.056^{b}	0.227^{a}	0.125	3.822**
Sport: Track and field	0.099^{bc}	0.000^{a}	0.000^{a}	3.638**
Sport: Football	0.099^{b}	0.296^{a}	0.208	3.767**
Job: Corporate health management	0.020^{c}	0.031^{c}	0.143^{ab}	2.670*
Job: Self-employed	0.157^{c}	0.188	0.429^{a}	3.466**
Job: Academic coordinator	0.014^{c}	0.000^{c}	0.083^{ab}	2.798*
Job: Student	0.352^{bc}	0.205^{a}	0.167^{a}	2.349*

Note: Displayed are the mean values; $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$; results of the LSD *post-hoc* test: ^{a}Significantly different from cluster 1; ^{b}Significantly different from cluster 2; ^{c}Significantly different from cluster 3.

Discussion

The first aim of this study was to examine the determinants of knowledge transfer from a sport science website which provides scientific knowledge in a less abstract and more digestible manner to sport practitioners. Two regression analyses were run to test the theoretical model of knowledge transfer proposed by Watson and Hewett (2006). The R^2s were slightly lower (Model 1) and higher (Model 2) than in their study, suggesting that the model is applicable to the present research context. Although it has originally been developed for knowledge transfer in organizations, it could also be applied to knowledge transfer beyond organizational boundaries. Nevertheless, differences in significant determinants of knowledge transfer point at differences between both contexts.

While ease of knowledge access had a significant positive effect on knowledge reuse in both studies, the perceived value of knowledge also affected knowledge reuse positively in Watson and Hewett (2006). This variable was insignificant in this study, presumably because of the high mean value ($M = 4.06$; $SD = 0.74$). With 2.40 ($SD = 0.70$), the perceived value was lower in Watson and Hewett (2006), indicating that the surveyed employees considered the knowledge provided to them by their organization as less valuable on average. While age was an insignificant predictor in Watson and Hewett (2006), older users reused knowledge provided by the website less frequently than their younger counterparts in this study. This difference suggests that knowledge reuse in organizations is more affected by the value employees assign to the knowledge, while frequent reuse of knowledge provided on a publicly available website is rather determined by age. Hence, the pattern that older individuals use the internet to a lesser extent (Van Deursen et al., 2015) also applies to knowledge reuse from a website.

Knowledge contribution was positively affected by knowledge reuse like in Watson and Hewett (2006) and also by job experience, which is similar to the positive association of tenure in their study. Hence, people who have accumulated more job-specific human capital were more likely to contribute knowledge not only within organizations, but also on publicly available websites. However, advancement within the organization and level within the company had significant positive effects in Watson and Hewett (2006), suggesting that organizational membership and commitment facilitate knowledge contribution. This was not the case for advancement within the job and one's level of sport science knowledge in this study, indicating lower commitment to the other users of the website and, hence, a lower willingness to contribute knowledge. Instead, while gender had no effect in Watson and Hewett (2006), males were more likely to contribute knowledge in this study. This finding might be explained by gender differences in general internet use patterns (Van Deursen et al., 2015) and the use of sport-related social media (Filo, Lock, & Karg, 2015).

The second aim was to categorize website users into different groups based on their level of knowledge transfer, resulting in three clusters. A high level of computer self-efficacy was only needed for frequent knowledge reuse, but not for knowledge contribution. This finding suggests the absence of technical barriers for knowledge contribution, meaning that other barriers must be present.

The socio-demographic factors indicate that males, older people, and those with more years of job experience were better represented among frequent contributors than among frequent users. Hence, respondents' gender, age, and job experience were important for the distinction of user groups, while educational level and formal qualifications did not play a role. This finding is interesting from a human capital perspective, which holds that individuals can invest in their human capital (i.e. knowledge, skills, abilities, etc.) through schooling institutions, on-the-job training, and participation in other activities (Becker, 1962). The present results suggest that individuals were more likely to share knowledge generated through on-the-job training (measured by job experience) than knowledge obtained by attending schooling institutions (measured by educational level) and participation in other activities (e.g. qualifications, certificates). This finding confirms existing research documenting that job experience represents an important facet of human capital in sport (Soebbing et al., 2016; Wicker et al., 2016). Likewise, older individuals (especially males) shared their human capital and knowledge more frequently, respectively, suggesting that not only job experience, but also life experience is critical to knowledge transfer.

While the three clusters did not differ regarding the topics of interest (e.g. nutrition, training, and recreation), they differed in terms of sports or conditional skills of interest people's current employment. Specifically, knowledge provided about endurance training sparked knowledge contribution, while knowledge about strength training was mainly consumed rather than shared. While knowledge about track and field was only accessed frequently but not shared, football-related knowledge was reused occasionally but shared. The infrequency of academic knowledge used by people involved in football is in line with existing research describing football as a

profession where people rely more on observation rather than on empirical data (Weimar & Wicker, 2017). Especially self-employed people and those employed as academic coordinators and in corporate health management seem to have generated knowledge in their jobs they consider worthy of sharing, presumably because they have applied such knowledge in their jobs already.

This study contributes to the literature in several ways. It is among the first to study knowledge transfer between science and practice using on a website, encompassing knowledge of the whole scientific discipline rather than only about a specific topic. The study also contributes to theory development in sport science by adapting an existing theoretical model to a context without organizational boundaries. Empirically, it adds to the literature by surveying several stakeholders of sport science and providing a typology of them based on their level of knowledge transfer.

The findings have implications for knowledge transfer from sport science to applied practice. Knowledge reuse of older people should be facilitated, for example by providing knowledge on topics relevant to this age group such as fitness and health (e.g. muscle hypertrophy, bone density, motor control). Given the overall low level of knowledge contribution, more opportunities and incentives that encourage users to contribute their knowledge more frequently are required, such as short questionnaires, interviews, or opinion barometers. Especially occasional female users of strength, fitness, and health content and knowledgeable users who are advanced in their job need to be incentivized to share their knowledge more frequently. One way could be to increase the commitment of these users to the website and to other users, for example by providing material or immaterial rewards for knowledge contribution.

This study has some limitations that represent avenues for future research. The first limitation is the low response rate and the resulting small sample size. Extending the database would be a fruitful perspective for future studies to see if the current findings are robust. Second, the regression models only explained 16.4% and 20.4% of the variation in the dependent variables, respectively, meaning that the findings need to be interpreted with caution. Further factors are needed to more fully explain knowledge transfer. One way to uncover such factors would be through qualitative analyses. Specifically, the barriers sport practitioners perceive with regard to knowledge transfer should be explored with a focus on females and older people. Third, the current study was limited to users of this particular website. Future research should study sport practitioners in general by building on the knowledge generated in this work.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Note

1. At the time of writing and data acquisition, all contents were publicly available, however, with prior registration to access material and watch videos, but without any fees or commercials.

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